OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

14.9 Signal positioning (deprecated)

The current ASAM OpenDRIVE only places signals at their physical location and uses signal `<reference>` and/or `<dependency>` elements to model, for example the concept of the interaction between stop lines and stop signs or traffic lights.

The ASAM OpenDRIVE Junction guideline describes interactivity between traffic lights and stop lines.

In previous versions of ASAM OpenDRIVE, a signal's position was identical with its validity and therefor should have been placed next to the road which it is valid for, enabling the application to identify the signals validity. This was called the logical position of a signal. The s-position of the signal described the position on the road where the signal takes effect.

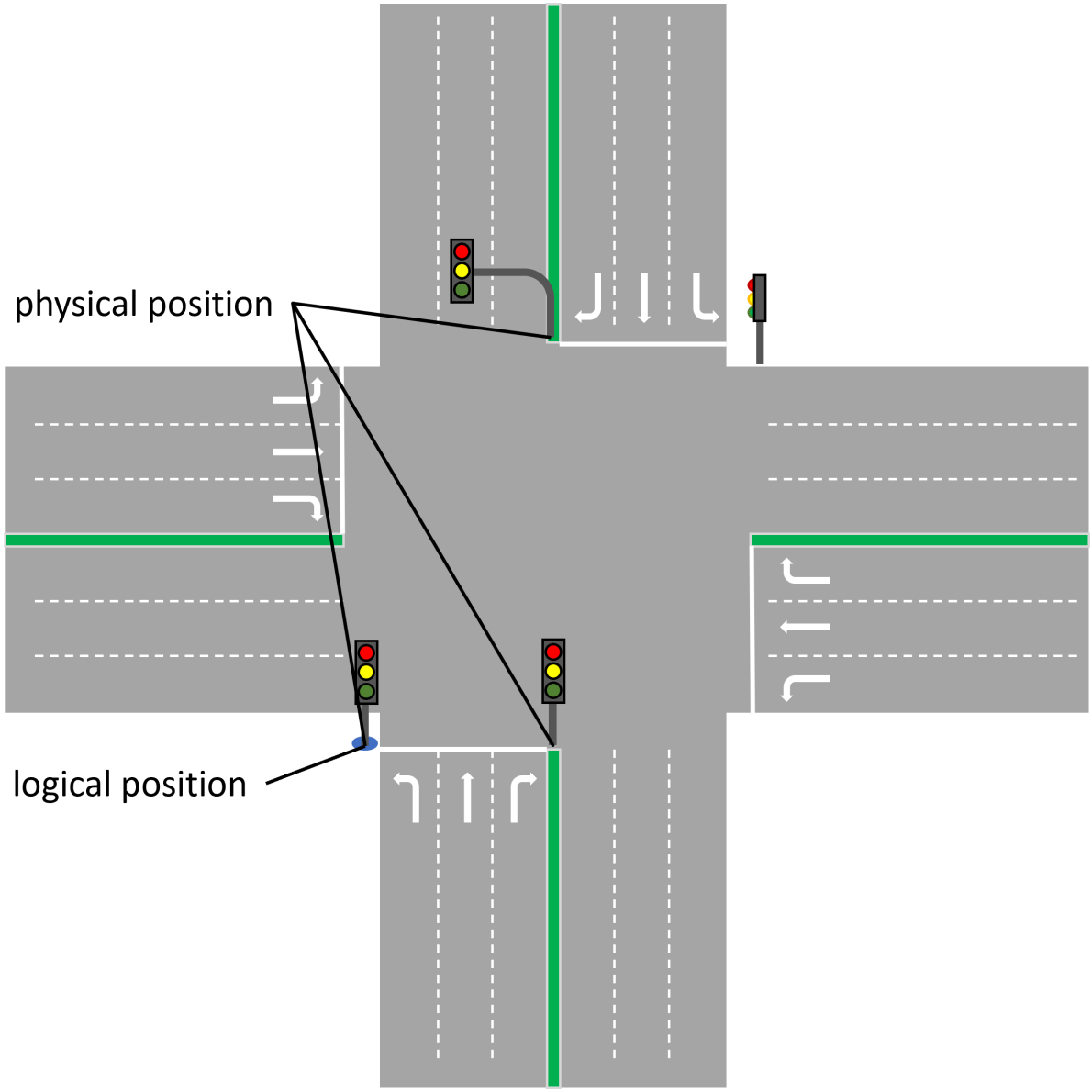

Figure 137. Junction with signals at physical and logical positions

Figure 137 shows how the physical and logical position of a signal could have differed in certain situations. ASAM OpenDRIVE offered two possibilities to describe the physical deviation of a signal. The possibilities were mutually exclusive. The positioning of the signal had no influence on its content.

- A signal may have been positioned at another physical position that is described with a road reference line coordinate system.
  A signal whose physical position deviated from its logical position was represented by the `<positionRoad>` element within the `<signal>` element. That means, the ID of the specified road was referenced, together with the s- and t-coordinates of the road.
  Examples were different positions of stop signs and stop lines.
- A signal may have been positioned at another physical position that was described with an inertial coordinate system. A signal whose physical position deviates from its logical position and was positioned using inertial coordinates was represented by the `<positionInertial>` element within the `<signal>` element.
  Inertial coordinates were used, for example, if the signal was not placed next to a road, but on the other side of the street or hanging over a junction.

Elements in UML model

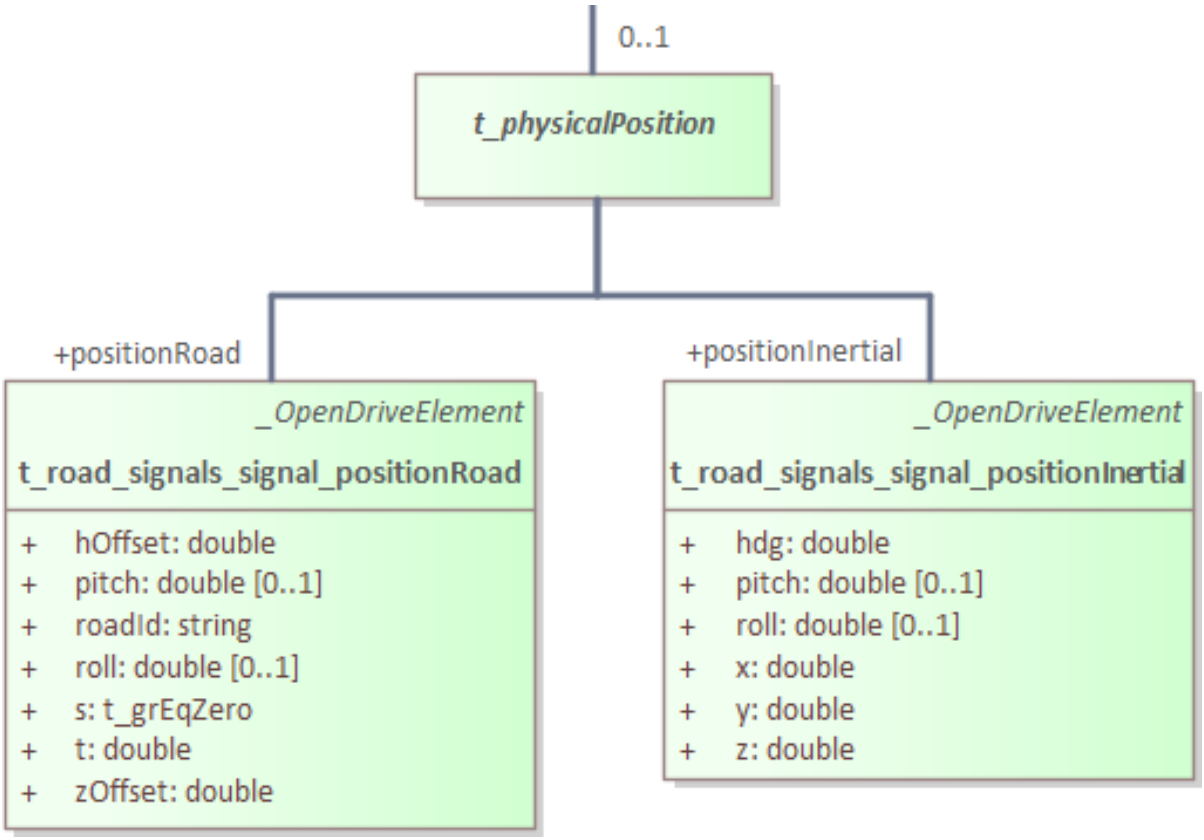

Figure 138. UML class diagram of the t_physicalPosition element in the Signals class

Figure 138 shows the UML class diagram of the t_physicalPosition element in the ASAM OpenDRIVE Signals class.

`<positionRoad>` element

In ASAM OpenDRIVE, a signal position using a referenced road is represented by the `<positionRoad>` element within the `<signal>` element.

```
UML class:  t_road_signals_signal_positionRoad
XML tag:    <positionRoad>
Deprecated: 1.8.0
```

Describes the reference point of the physical position road coordinates in cases where it deviates from the logical position. Defines the position on the road.

Table 140. Attributes of the <positionRoad> element

Name	Type	Use	Unit	Description
`hOffset`	double	required	rad	Heading offset of the signal (relative to @orientation)
`pitch`	double	optional	rad	Pitch angle of the signal after applying hOffset, relative to the inertial system (x'y'-plane)
`roadId`	string	required		Unique ID of the referenced road
`roll`	double	optional	rad	Roll angle of the signal after applying hOffset and pitch, relative to the inertial system (x"y"-plane)
`s`	t_grEqZero	required	m	s-coordinate
`t`	double	required	m	t-coordinate
`zOffset`	double	required	m	z offset from road level to bottom edge of the signal

`<positionInertial>` element

In ASAM OpenDRIVE, a signal position using inertial coordinates is represented by the `<positionInertial>` element within the `<signal>` element.

```
UML class: t_road_signals_signal_positionInertial
XML tag:   <positionInertial>
```

Describes the reference point of the physical position in inertial coordinates in cases where it deviates from the logical position. Defines the inertial position.

Table 141. Attributes of the <positionInertial> element

Name	Type	Use	Unit	Description
`hdg`	double	required	rad	Heading of the signal, relative to the inertial system
`pitch`	double	optional	rad	Pitch angle of the signal after applying heading, relative to the inertial system (x'y'-plane)
`roll`	double	optional	rad	Roll angle of the signal after applying heading and pitch, relative to the inertial system (x"y"-plane)
`x`	double	required	m	x-coordinate
`y`	double	required	m	y-coordinate
`z`	double	required	m	z-coordinate

XML example

- UC_LHT_Complex-TrafficLights.xodr

Rules

The following rules apply to signal positioning:

- Signals should be placed next to the road for which they are valid.
- The physical position of signals may deviate from their logical position.

Related topics

- Section 14.1, "Introduction to signals"
- Section 14.2, "Lane validity for signals"
- Section 14.3, "Signal dependency"
- Section 14.4, "Signal reference"